Appendix C

CompTIA Network+ Practice Exam

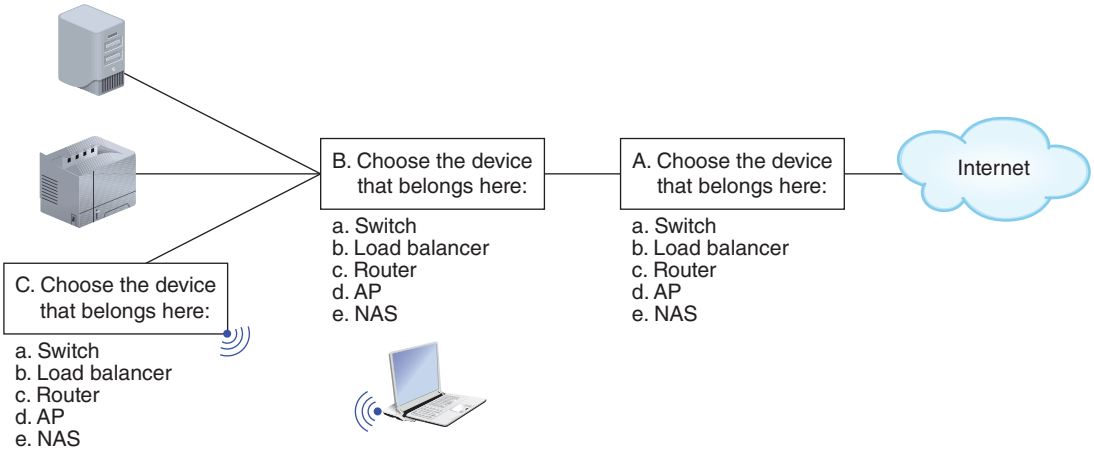
The following practice exam contains questions that are similar in content and format to the questions you will encounter on CompTIA's Network+ N10-009 certification exam, released in 2024. This practice exam consists of 90 questions. Some questions have more than one correct answer. The number of questions from each domain reflects the weighting that CompTIA assigned to these domains in its exam objectives. To simulate taking the CompTIA Network+ certification exam, allow yourself 90 minutes to answer all the questions.

Note C-1 | The CompTIA Network+ exam begins with two to five PBQs (performance-based questions). These questions require different types of interactions than do multiple choice questions. For example, you might need to make selections from drop-down lists or reposition items on the screen. And often, you must collect information not only from the question's scenario but also from the other items on the screen. For example, you might need to click a device's image to display its configuration information.

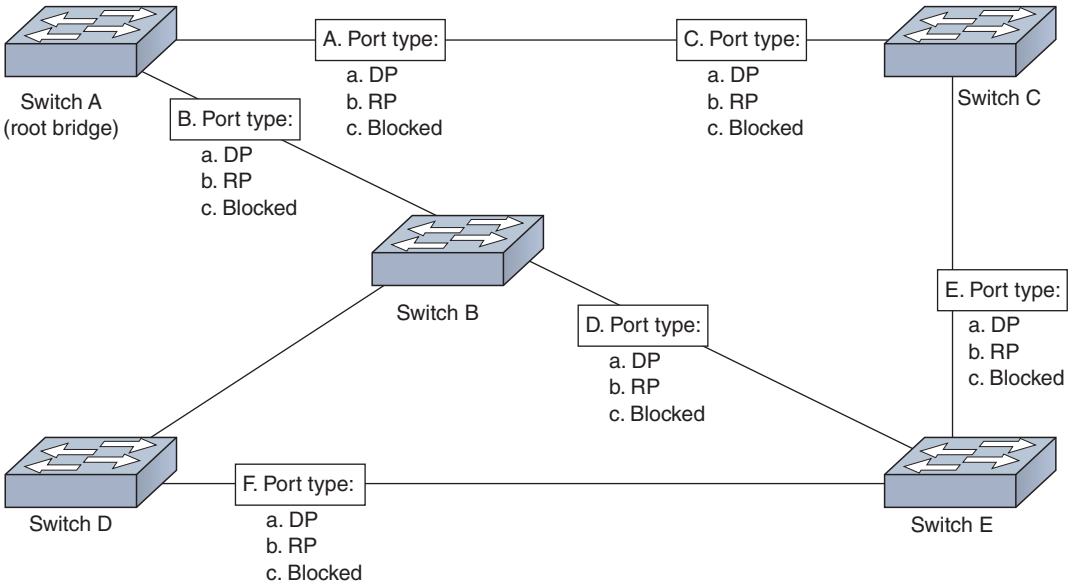
This practice exam begins with four PBQs that simulate the types of PBQs you will encounter on the CompTIA Network+ exam while also being compatible with static text (such as when you're reading a textbook). The PBQs on the real exam will require significant amounts of interactivity. Although this interactivity cannot be duplicated in a textbook, the sample PBQs here will help you imagine the kinds of PBQs you will find on the real exam.

Notice that PBQs ask about information you've learned in this course. However, it presents the scenario and the information in a way that helps you think differently about and apply the information you've learned.

1. Examine the network diagram and select the device that belongs in each location.



2. Identify the STP port type for each labeled interface.



3. You're setting up a new home router for a customer. Because the customer's existing home network will be used for their televisions, smartphones, and laptops, the new Wi-Fi network will only need to support IoT devices, such as security cameras, smoke detectors, light bulbs, and a thermostat. Additionally, many of these devices will be positioned relatively far from the home router. From the provided options, configure the home router with optimal settings to support and secure the IoT network.

Wi-Fi Home Router **Model: WFR-1051**

A. SSID:
a. WFR-1051
b. Yellow-House
c. Davis-Family
d. IOT-network
e. Blue-Box

B. Wi-fi password:
a. admin
b. t56-98#@jwvd73^i
c. D@v1s-F@m1ly
d. P@$$w0rd
e. yellowmonthcamel

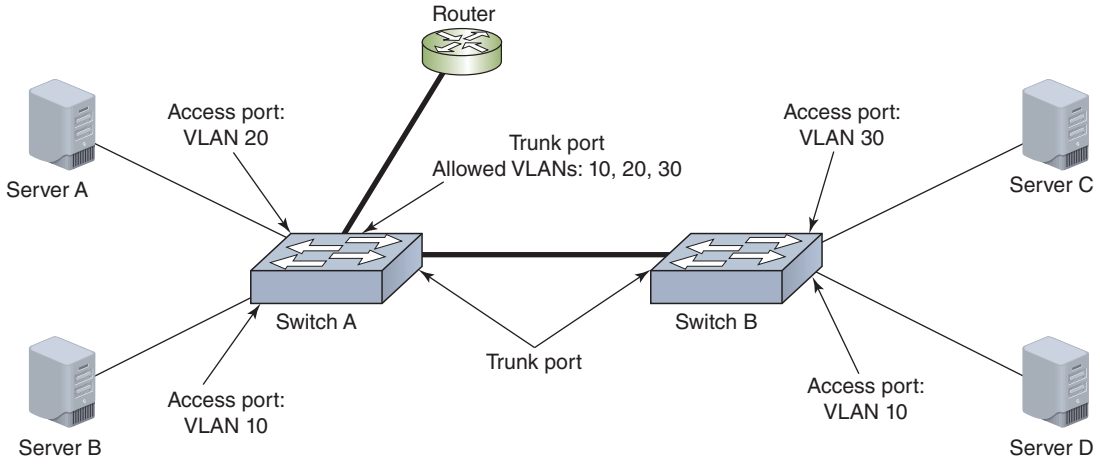
C. Frequency band:
a. 2.4 GHz
b. 5 GHz
c. 6 GHz

D. Channel:
a. 1
b. 6
c. 11
d. 15
e. auto

E. Security:
a. WEP
b. WPA
c. WPA2-PSK
d. WPA2-Enterprise
e. WPA3-Personal
f. WPA3-Enterprise

F. WPS:
a. Enable
b. Disable

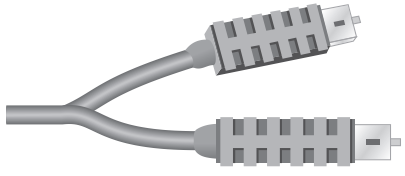
4. Examine the network diagram illustrating two switches and a router supporting three VLANs: 10, 20, and 30. All the servers are able to communicate across VLANs; however, server A should be isolated from the others and should not be allowed to communicate across VLANs. Which device's configuration should be changed to meet this requirement?



a. Router
b. Server A
c. Server C
d. Switch A
e. Switch B

5. To ensure that your private network is always protected, you decide to install three redundant firewalls. Which of the following would allow you to assign the same IP address to all three?
a. SMTP
b. CARP
c. SNMPv3
d. IMAP
e. NTP

6. What Ethernet standard could use the connector illustrated here?



a. 1000BASE-LX
b. 10GBASE-T
c. 100BASE-TX
d. 1000BASE-T
e. 10GBASE-T

7. While troubleshooting a workstation connectivity problem, you enter the following command: `ping 127.0.0.1`. The response indicates the test failed. What can you determine about that workstation?
a. Its network cable is faulty or not connected to the wall jack.
b. Its TCP/IP stack is not installed properly.
c. It has been prevented from transmitting data past the default gateway.
d. Its DHCP settings are incorrect.
e. Its DNS name server specification is incorrect.

8. You have been asked to help improve network performance on a store's small office network. The network relies on two switches, two access points, and a router to connect its 18 employees to the Internet and other store locations. You decide to determine what type of traffic the network currently handles. In particular, you're interested in the volume of unnecessary broadcast traffic that might be bogging down shared segments. Which of the following tools will help you identify the percentage of traffic made up of broadcasts?
   a. Port scanner
   b. OTDR
   c. Protocol analyzer
   d. Multimeter
   e. Cable tester

9. Which of the following standards describes a security technique, often used on wireless networks, in which a port is prevented from receiving traffic until the transmitter's credentials are verified by an authentication server?
   a. EAPoL
   b. SSH
   c. RADIUS
   d. Kerberos
   e. CCMP

10. Which of the following ports would be used during a domain name lookup?
   a. 22
   b. 23
   c. 53
   d. 110
   e. 443

11. You are configuring a connection from a Windows server to a switch, and you want to make sure the connection doesn't fail or become overwhelmed by heavy traffic. Which of the following techniques would help you achieve both aims?
   a. Multipathing
   b. CARP
   c. Clustering
   d. Trunking
   e. NIC teaming

12. As a network admin, you have decided to install additional physical security to the main office's data room. Due to the sensitivity of the data held in this room, you decide it's critical to ensure two-factor authentication before granting anyone access to the room. You already have a lock on the door. Which of the following physical security measures would provide 2FA?
   a. Smart card
   b. Fingerprint scanner
   c. Keychain fob
   d. Video surveillance
   e. Proximity card

13. You have installed and configured two virtual web servers and a virtual mail server on a physical server. What networking mode will you assign to each server's vNIC to ensure that the virtual machines' clients on the Internet can access the virtual machines?
   a. NAT
   b. Bridged
   c. Host-only
   d. Internal
   e. Grouped

14. When using NAT, how does an IP gateway ensure that outgoing traffic can traverse public networks?
    a. It modifies each outgoing frame's Type field to indicate that the transmission is destined for a public network.
    b. It assigns each outgoing packet a masked ID via the Options field.
    c. It replaces each outgoing packet's Source address field with a public IP address.
    d. It interprets the contents of outgoing packets to ensure that they contain no client-identifying information.
    e. It modifies the frame length to create uniformly sized frames, which are required for public network transmission.

15. Which of these authentication techniques only encrypts the password when transmitting sign-in credentials?
    a. RADIUS
    b. TACACS+
    c. Kerberos
    d. Single sign-on
    e. Local authentication

16. You are a networking technician in a radiology clinic where physicians use the network to transmit and store patients' diagnostic results. Shortly after a new wing, which contains X-ray and MRI (magnetic resonance imaging) machines, is added to the building, computers in that area begin having intermittent problems saving data to the file server. After you have gathered information, identified the symptoms, questioned users, and determined what has changed, what is your next step in troubleshooting this problem?
    a. Establish a plan of action to resolve the problem.
    b. Escalate the problem.
    c. Document findings, actions, and outcomes.
    d. Establish a theory of probable cause.
    e. Implement the solution.

17. The software on a firewall you recently installed on your network examines each incoming packet. It blocks or allows traffic based on a set of criteria—including source IP address, destination ports, and TCP flags—and it does not evaluate traffic in relation to other traffic. What type of system is this? Choose TWO.
    a. Host-based firewall
    b. Stateful firewall
    c. Stateless firewall
    d. Packet-filtering firewall
    e. Application layer firewall

18. Suppose you are creating six subnets on a network that leases a group of class C IPv4 addresses to desktop computers, servers, and printers. Because the network rarely needs to connect mobile devices, what DHCP configuration will minimize the number of times a client device must request an updated IP address?
    a. Lease time
    b. IP reservation
    c. Scope
    d. IP exclusion
    e. Default gateway

19. What would the command `route del default gw 192.168.5.1 eth1` accomplish on your Linux workstation?
    a. Delete the default gateway's route to the host whose IP address is 192.168.5.1.
    b. Remove the assignment of IP address 192.168.5.1 from the eth1 interface.
    c. Remove the workstation's route to the default gateway whose IP address is 192.168.5.1.
    d. Add a route from the workstation to the default gateway whose IP address is 192.168.5.1.
    e. Remove the designation of default gateway, but keep the route for the host whose IP address is 192.168.5.1.

20. From your laptop, you need to remote into a switch to make some configuration changes. Which transport layer protocol and TCP/IP port should you open in Windows Firewall to make this work using Telnet?
    a. UDP, port 23
    b. TCP, port 23
    c. UDP, port 22
    d. TCP, port 22
    e. UDP, port 21

21. Recently, your company's WAN experienced a disabling DDoS attack. Which of the following devices could detect such an attack and prevent it from affecting your network in the future?
    a. A honeypot
    b. SIEM
    c. HIPS
    d. HIDS
    e. NIPS

22. Which of the following routing protocols offer fast convergence times and can be used on both core and edge routers? Choose TWO.
    a. RIPv2
    b. IS-IS
    c. OSPF
    d. BGP
    e. EIGRP

23. A friend calls you for help with their home office Internet connection. They are using an 802.11ac wireless router connected to a DSL modem. The router's private IP address is 192.168.1.1, and it has been assigned an Internet routable IP address of 76.83.124.35. Your friend cannot connect to any resources on the Internet using their new Windows workstation. You ask your friend to run the `ipconfig` command and read the results to you. Your friend says their workstation's IP address is 192.168.1.3, the subnet mask is 255.255.255.0, and the default gateway address is 192.168.1.10. What do you advise them to do next?
    a. Display their DNS information.
    b. Change their gateway address.
    c. Change their subnet mask.
    d. Try pinging the loopback address.
    e. Use the `tracert` command to contact the access point/router.

24. Your organization contracts with a cloud service provider to store some backup data. The company promises 99.999 percent uptime. If it lives up to its claims, what is the maximum number of minutes each year you can expect your data to be unavailable?
    a. Approximately 448 minutes
    b. Approximately 199 minutes
    c. Approximately 52 minutes
    d. Approximately 14 minutes
    e. Approximately 5 minutes

25. In a spine-and-leaf architecture, what type of switch is typically connected to a spine layer switch?
    a. Core switch
    b. Distribution switch
    c. Access switch
    d. Leaf switch
    e. Spine switch

26. What is the default subnet mask for the IP address 154.13.44.87?
    a. 255.255.255.255
    b. 255.255.255.0
    c. 255.255.0.0
    d. 255.0.0.0
    e. 0.0.0.0

27. Your CFO has approved installing new backbone cabling on your school's campus. One of the buildings is particularly far away from the others—nearly a kilometer. Which Ethernet standard will reach the distant building without the use of a repeater?
    a. 10GBASE-T
    b. 1000BASE-LX
    c. 1000BASE-SX
    d. 10GBASE-SR
    e. 1000BASE-T

28. Which of the following is often used to secure data traveling over VPNs that use L2TP?
    a. OpenVPN
    b. GRE
    c. Kerberos
    d. SSH
    e. IPsec

29. You are a support technician working in a data closet in a remote office. You suspect that a connectivity problem is related to a broken RJ-45 plug on a patch cable that connects a switch to a patch panel. You need to replace that connector, but you forgot to bring an extra patch cable. You decide to install a new RJ-45 connector to replace the broken connector. Which tools must you have to successfully accomplish this task? Choose TWO.
    a. Punchdown tool
    b. Cable crimper
    c. Wire cutter
    d. Cable tester
    e. Multimeter

30. You have purchased an outdoor access point capable of exchanging data via the 802.11n or 802.11ac wireless standard. According to these standards, what is the maximum distance, in meters, from the access point that wireless clients can travel and still reliably exchange data with the access point if there are no obstacles between them?
    a. 20
    b. 70
    c. 100
    d. 250
    e. 450

31. Your friend's printer isn't printing the document they just sent it. In what order should you perform the listed steps?
    a. Follow the OSI model from bottom to top to check possible causes, send a new document to the printer, and determine if anything has changed on their network.
    b. Send a new document to the printer, follow the OSI model from bottom to top to check possible causes, and ask your friend when the problem started.
    c. Take notes on the outcome, send a new document to the printer, and determine if anything has changed on their network.
    d. Determine if anything has changed on your friend's network, follow the OSI model from bottom to top to check possible causes, and send a new document to the printer.
    e. Determine if anything has changed on your friend's network, take notes on the outcome, and send a new document to the printer.

32. A CEO fires their administrative assistant after the assistant was caught stealing company funds. Over the weekend, the administrative assistant hacks into the CEO's private email account and steals some personal data. What type of attack did the former employee most likely use to accomplish this exploit?
    a. Brute force attack
    b. War driving
    c. Logic bomb
    d. Deauth attack
    e. On-path attack

33. What is the network ID for a class C network that contains the group of IP addresses from 194.73.44.10 through 194.73.44.254?
    a. 194.73.44.0
    b. 194.73.44.1
    c. 194.73.0.0
    d. 194.73.44.255
    e. 194.1.1.1

34. Your organization is reassessing its WAN connections to determine how much more bandwidth it will need to purchase in the next two years. As a network administrator, which of the following data can you share that will help management make the best decision?
    a. Wiring diagram
    b. Baseline
    c. Logical topology
    d. Syslog
    e. Change management documentation

35. You are creating a new Linux server as a virtual machine on your Windows workstation. Which of the following commands will tell you the IP address that is assigned to your virtual server?
    a. `ipconfig /all` at the Windows workstation's command prompt
    b. `ifconfig -a` at the Linux server's shell prompt
    c. `iptables` at the Linux server's shell prompt
    d. `ping` at the Linux server's shell prompt
    e. `ipconfig /all` at the Linux server's shell prompt

36. Which of the following requirements provide 2FA? Choose TWO.
    a. Iris pattern and typing pattern
    b. Password and name of first elementary school
    c. Location in a secured closet
    d. Smart card and key fob
    e. Fingerprint and name of first elementary school

37. You are rearranging nodes on your Gigabit Ethernet network. Due to a necessarily hasty expansion, you have decided to supply power to a wireless router in a makeshift data room using PoE. Which of the following is likely the cheapest cabling you could use to connect this wireless router to the network's backbone?
    a. RG-6
    b. RG-59
    c. Cat 5e
    d. SMF
    e. Cat 6

38. As you're setting up APs in your client's office space, you want to ensure that all work areas and the meeting room have adequate access to the network. What tool will give you the information you need?
    a. Geofencing
    b. Packet sniffer
    c. Bandwidth speed tester
    d. Wi-Fi analyzer
    e. Toner probe

39. Which of the following protocols encapsulates data for transmission over VPNs?
    a. SFTP
    b. L2TP
    c. RDP
    d. TCP
    e. TACACS+

40. Which of the following is a valid MAC address?
    a. C3:00:50:00:FF:FF
    b. 153.101.24.3
    c. ::9F53
    d. FE80::32:1CA3:B0E2
    e. D0:00:00:00

41. Which of the following services would be most important to disable on a Windows workstation? Choose TWO.
    a. SSH
    b. DHCP
    c. Telnet
    d. FTP
    e. RDP

42. It's Friday night, and you have just settled in with some hot cocoa and popcorn to watch one of your favorite movies on Netflix. Five minutes into the movie, you realize you're getting more stressed than relaxed, and then you realize the problem is that the movie keeps lagging, buffering, and skipping. What transmission flaw is probably the source of the problem?
    a. Crosstalk
    b. Jitter
    c. EMI
    d. Latency
    e. Attenuation

43. Where might you find publicized information about a backdoor to an application?
    a. CIA
    b. EAP
    c. SIEM
    d. CVE
    e. SSO

44. As you type in your PIN at the ATM, what risk can you protect yourself from by covering the keypad as you type?
- **a.** Piggybacking
- **b.** Phishing
- **c.** Tailgating
- **d.** Shoulder surfing
- **e.** Spoofing

45. After running the `show interface` command on a router, which field in the output indicates a damaged but minimally functioning cable?
- **a.** Giants
- **b.** BW
- **c.** CRC errors
- **d.** Runts
- **e.** Link state

46. You suspect that a machine on your network with the host name PRTSRV is issuing excessive broadcast traffic. What command can you use to determine this host's IP address?
- **a.** `netstat PRTSRV`
- **b.** `ipconfig PRTSRV`
- **c.** `ping -4 PRTSRV`
- **d.** `ifconfig -a PRTSRV`
- **e.** `ip address show PRTSRV`

47. You're installing a network management system and need to install software on various devices. What software must be installed on a managed device, such as a router?
- **a.** SIEM
- **b.** OID
- **c.** NMS
- **d.** Agent
- **e.** Syslog

48. You work for a small ISP. Several of your customers have called to complain about slow responses from a popular website. You suspect that network congestion is at fault. Which TCP/IP utility would help you determine where the congestion is occurring?
- **a.** Tracert
- **b.** Nslookup
- **c.** ARP
- **d.** FTP
- **e.** Telnet

49. On an IPv6 network, which of the following is a host computer's loopback address?
- **a.** 1.0.0.1
- **b.** ::1/128
- **c.** FE80::1
- **d.** 127::1
- **e.** 127.0.0.1

50. In the process of troubleshooting an intermittent performance problem with your network's Internet connection, you attempt to run a `tracert` test to microsoft.com. The `tracert` response displays the first 12 hops in the route but then presents several "Request timed out" messages in a row. What is the most likely reason for this result?
- **a.** Your network's ISP is experiencing connectivity problems.
- **b.** Microsoft's network is bounded by firewalls that do not accept incoming ICMP traffic.
- **c.** The Internet backbone is experiencing traffic congestion.

d. Your client's TCP/IP service limits the `tracert` command to a maximum of 12 hops.
e. Your IP gateway failed while you were attempting the `tracert` test.

51. You are setting up a new Windows client to connect with your LAN, which relies on DHCP. Which of the following must you do to ensure that the client obtains correct TCP/IP information via DHCP?
    a. Make certain the client's computer name and host name are identical.
    b. Enter the client's MAC address in the DHCP server's ARP table.
    c. Nothing; in Windows, the DHCP option is selected by default.
    d. Enter the DHCP server address in the Windows TCP/IP configuration.
    e. Enter a default gateway address in the Windows TCP/IP configuration.

52. Due to popular demand from employees who need to roam from one floor of your office building to another, you are expanding your wireless network. You want to ensure that mobile users enjoy uninterrupted network connectivity without having to reconfigure their workstations' wireless network connection settings as they travel throughout the office space. Which of the following variables must you configure on your new access points to match the settings on existing access points?
    a. Administrator password
    b. Scanning rate
    c. IP address
    d. RSSI
    e. ESSID

53. You have installed a protocol analyzer on your laptop and connected the laptop to a switch on your network's backbone. You want to monitor all traffic on a specific VLAN. Which feature must you configure on the switch to make this work?
    a. Trunking
    b. Port mirroring
    c. Looping
    d. Spanning Tree Protocol
    e. Caching

54. Which of the following does NOT accurately describe TACACS+ in comparison to RADIUS?
    a. TACACS+ relies on TCP, not UDP, at the transport layer.
    b. TACACS+ is used to authenticate to a router or switch.
    c. TACACS+ encrypts all information transmitted for AAA rather than just the password.
    d. TACACS+ was developed for proprietary use on Cisco products.
    e. TACACS+ operates as a software application on a remote access server.

55. You are part of a team participating in a posture assessment of your company's LAN. Which of the following tools or strategies will help you gain a broad understanding of your network's security vulnerabilities?
    a. MIB
    b. Geofencing
    c. Nmap
    d. DHCP snooping
    e. CCMP

56. Which of the following devices can be used to separate broadcast domains? Choose TWO.
    a. Router
    b. Switch
    c. Access point
    d. Repeater
    e. Hub

57. About a year ago, you purchased and installed a new router at one of your company's branch offices, and you have had nothing but problems with the router since. Today you called the vendor for the third time to have them send a technician to repair the router. They estimate the repair will be completed by the end of business tomorrow. In the meantime, you have decided that you would prefer to replace the router the next time it fails. What factor should you carefully investigate when shopping for a replacement, so you won't face this same scenario with the next router?
   a. MTBF
   b. MTTR
   c. UPS
   d. SLA
   e. MIB

58. A 500-watt transmitted signal arrives at the receiver at 250 watts. What is the signal's dB loss?
   a. 250 dB loss
   b. 3 dB loss
   c. 1,000 dB loss
   d. 6 dB loss
   e. 1 dB loss

59. Your study partner is trying to access the Internet on their laptop, and it's not working. You offer to help, and you use their browser to try to go to google.com. That doesn't work. In a Command Prompt window, you ping google.com, which doesn't work either. You then ping 8.8.8.8, and it still doesn't work. You try to ping the network's gateway, and it fails. You ping your own computer, which is logged onto the same network, and it works. What is the most likely diagnosis of this problem?
   a. Exhausted DHCP scope
   b. Incorrect gateway
   c. Incorrect DNS server
   d. Duplicate IP address
   e. Incorrect subnet mask

60. You work for a small fashion design firm. Because of a video that recently went viral, your company has received national recognition. Within a few days, your web server crashes. What kind of attack most likely caused the crash?
   a. Phishing
   b. Friendly DoS
   c. Spoofing
   d. Ransomware
   e. Logic bomb

61. You have just installed Linux on an old laptop to bring it new life. Which of the following applications will provide the most protection for the computer when you connect it to the Internet?
   a. Defender
   b. Honeypot
   c. Tcpdump
   d. Wireshark
   e. Iptables

62. You are the network administrator for a large university whose network currently contains nearly 10,000 workstations, over 80 routers, 250 switches, and 500 printers. You are researching a proposal to upgrade the routers and switches on your network to primarily use multilayer switches instead. At the same time, you want to improve the management of your network devices. Which of the following protocols will help automate network management across all the new devices?
   a. SNMP
   b. TFTP
   c. SMTP

d. NTP
e. ICMP

63. While configuring your new SOHO router, you're given several security options. Which one would provide the most secure encryption of your transmissions?
a. PSK
b. AES
c. WEP
d. TKIP
e. RC4

64. You have connected to your bank's home page. Its URL begins with https://. Based on this information, what type of security can you assume the bank employs for clients receiving and transmitting data to and from its web server?
a. SSO
b. TLS
c. IPsec
d. L2TP
e. ACL

65. Which of the following routing protocols has the poorest convergence time?
a. EIGRP
b. RIP
c. BGP
d. OSPF
e. IS-IS

66. You need to gather some information about the traffic being broadcast and received on a Linux workstation. Unfortunately, you don't have administrative permissions to install Wireshark. Which command can you use instead?
a. `tcpdump`
b. `telnet`
c. `traceroute`
d. `dig`
e. `nslookup`

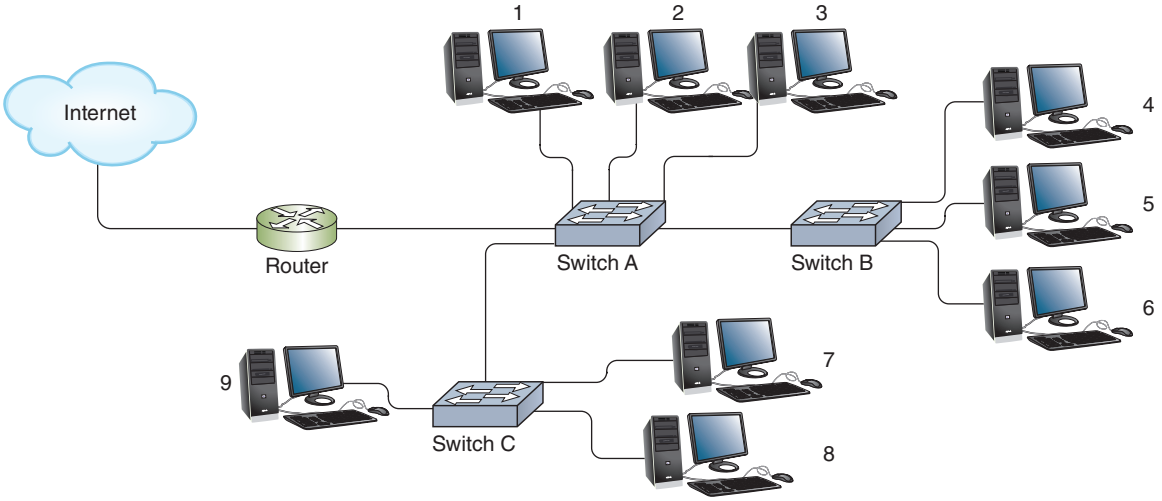
67. To provide redundancy, you need to set up three links from a critical server to two redundant switches, both of which operate on the same subnet. Two connections from the server go to one switch, and the third link connects the other switch to the server. The server is running Windows Server. What feature should you configure on the server to make these redundant links work together for load balancing and failover protection?
a. Port aggregation
b. Flow control
c. Server cluster
d. NIC teaming
e. Virtual IP address

68. A Windows workstation is configured to use DHCP, but it cannot find a DHCP server. You run `ipconfig` in PowerShell. Which of the following IP addresses is most likely reported by Windows?
a. 129.0.0.1
b. 255.255.255.0
c. 192.168.0.1
d. 169.254.1.120
e. 172.16.2.18

69. When used side-by-side with Kerberos, what service does LDAP provide Active Directory?

a. Authorization
b. Accounting
c. Encryption
d. Auditing
e. Authentication

70. In the following figure, if switch A suffers a failure, how will this failure affect nodes 1 through 3?



a. They will be unable to access the Internet or nodes 7 through 9.
b. They will be unable to access the Internet or any other nodes on the LAN.
c. They will only be unable to access the Internet.
d. They will be unable to access nodes 4 through 9.
e. Their connectivity will not be affected.

71. A file server on your network is running Ubuntu Server. You need to remote into it to make a backup of the server's configuration settings. Which of the following utilities will give you secure access to the server to make your backup?

a. RDP
b. SSH
c. SSL
d. SFTP
e. Telnet

72. You want to add five VMs on a server to the Staff VLAN at your office and two VMs to the Sales VLAN. Which of the following must your host machine's NIC support?

a. NAT
b. Channel bonding
c. 802.1Q tagging
d. Trunking
e. OSPF

73. A regional bank manager asks you to help with an urgent network problem. Because of a sudden and severe network performance decline, the manager worries that the bank's network might be suffering a DoS attack. Viewing which of the following types of network documentation would probably give you the quickest insight into what's causing this problem?

a. Wiring diagram
b. Firewall log
c. Logical network diagram

d. The main file server's system log
e. Physical network diagram

74. While troubleshooting a computer's connection to the network, you enter a command at the Command Prompt and see the following output. Which command did you enter?

```
Default Server:  google-public-dns-a.google.com
Address:  8.8.8.8

>
```

a. `nslookup`
b. `netstat`
c. `arp -a`
d. `tracert google.com`
e. `ping 8.8.8.8`

75. On which of the following devices would you NOT change the default administrative username and password when adding each device to your network?
a. IP security camera
b. Router
c. Programmable thermostat
d. Unmanaged switch
e. Printer

76. Which of the following transmission media is most resistant to interference?
a. Twinaxial cable
b. Fiber-optic cable
c. STP cable
d. 802.11ac transmissions
e. Coaxial cable

77. You're setting up a wireless network for a coffee shop and want to provide some liability protection for the shop owners when customers use their Wi-Fi Internet access. Which of the following tools will provide this protection while still allowing customers convenient access to the Internet?
a. ACL
b. WPA2
c. MAC filter
d. EAPoL
e. Captive portal

78. While troubleshooting a printer problem, you figured out the printer's static IP address had not been reserved on the DHCP server and so was mistakenly issued to a workstation. This caused a conflict and prevented users from accessing the printer. You have now configured the reservation in DHCP. What should you do next?
a. Document the problem and your solution in your company's knowledge base.
b. Close the service ticket.
c. Report your findings to the network admin.
d. Check that the printer is plugged into the wall outlet.
e. Confirm that users can now print successfully.

79. How does STP prevent or stop broadcast storms?
    a. It examines the source IP address field in each broadcast packet and temporarily blocks traffic from that address.
    b. It enables routers to choose one set of best paths and ensures that alternate paths are used only when the best paths are obstructed.
    c. It enables switches to calculate paths that avoid potential loops and artificially blocks the links that would complete a loop.
    d. It allows switches to monitor IP address assignments on a network.
    e. It lets switches restrict each interface's MAC address table to a single MAC address.

80. What is the function of protocols and services at the network layer of the OSI model?
    a. To manage the flow of communications in both directions during a session
    b. To add segmentation and assembly information
    c. To encode and encrypt data
    d. To add logical addresses and properly route data
    e. To apply electrical pulses to the wire

81. You have created a new web server on a computer running the Linux operating system. Some of the modules aren't loading correctly, and services are encountering errors. Which of the following applications should you check for information on these errors?
    a. Event Viewer
    b. IDS
    c. Packet sniffer
    d. NetFlow
    e. Syslog

82. As you're creating a new website for your organization, you need to add new DNS records for the website and validate your DNS server's information so other DNS servers will accept the updates. Which standard will meet this requirement?
    a. DNSSEC (DNS Security Extensions)
    b. CNAME (canonical name)
    c. NS (nameserver)
    d. DoH (DNS over HTTPS)
    e. DoT (DNS over TLS)

83. Suppose your Windows laptop's wireless network adapter is configured to use the 802.11n wireless networking standard. Also, suppose a café you visit has a popular model of an 802.11ac access point. Assuming you have the correct SSID and logon credentials, what will most likely happen when you attempt to associate with the café's wireless network?
    a. Your wireless networking client will be able to see the access point but will be unable to associate with it.
    b. Your wireless networking client will not be able to see the access point.
    c. Your wireless networking client will be able to see the access point and attempt to associate with it, but the incompatible frequencies will prevent successful association.
    d. Your wireless networking client will be able to see the access point and attempt to associate with it, but the incompatible security techniques will prevent successful association.
    e. Your wireless networking client will be able to see the access point and successfully associate with it.

84. Which of the following commands lists active TCP/IP connections on both Windows and Linux computers?
    a. `nslookup`
    b. `netstat`
    c. `hostname`
    d. `ipconfig`
    e. `route`

85. You have just rearranged the access points on your small office network. Now a group of employees complains that they cannot reliably get their workstations to connect with a new 802.11ac access point. You have confirmed that the workstations are using the correct SSID, security type, and passphrase. You have also confirmed the access point is turned on and functioning properly because when you stand in the computer room where it's located, you can connect to the access point from your smartphone. What should you do next to confirm the most likely problem?
   a. Check authentication logs for failed logon attempts.
   b. Check a heat map for access point signal strength.
   c. Check workstation configurations for Wi-Fi standard used.
   d. Check the network server's interface logs for bottlenecked traffic.
   e. Check workstation configurations for DHCP assignments.

86. Which protocol supports recovery of a router config file?
   a. TFTP
   b. UPS
   c. SSH
   d. SNMP
   e. OID

87. Which of the following is a likely reason for using VLANs?
   a. To facilitate easier migration from IPv4 to IPv6 addressing
   b. To enable DHCP on a network
   c. To limit broadcast domains
   d. To reduce the likelihood for user error when modifying TCP/IP properties
   e. To reduce the number of routing table entries

88. Which of the following wireless security techniques uses both RADIUS and AES?
   a. WPA2-Enterprise
   b. WPA
   c. WEP
   d. WPA-Enterprise
   e. WPA2

89. Your company is experiencing a growth spurt and is ready to invest in a more sophisticated disaster recovery plan. Currently the backup plan consists of a few spare computers in a storage closet, and data on the servers is duplicated weekly to an off-site backup service. The company owners have committed to acquiring additional servers to duplicate critical servers in their current network, and they want the servers to be configured identically to the servers now in use. The new servers will be stored at an off-site data center and updated every time the on-site servers are updated. What type of disaster recovery site is your company creating?
   a. Hot site
   b. Ambient site
   c. Cloud site
   d. Warm site
   e. Differential site

90. A virtual switch includes several virtual ports, each of which can be considered a _____.
   a. virtual bridge
   b. virtual router
   c. virtual gateway
   d. virtual hub
   e. virtual repeater